STUDY GUIDE

Discrete Math Essentials

Notes, definitions, worked examples and exam patterns.

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1. Logic and Proofs
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Logic and Proofs

Contrapositive. $p \to q$ is logically equivalent to $\neg q \to \neg p$.

Claim. There are infinitely many primes.

Proof. Suppose $p_1, \ldots, p_k$ are all the primes and let $N = p_1 p_2 \cdots p_k + 1$. No p_i divides N, but $N > 1$ has a prime factor. Contradiction.

Induction

$$\sum_{i=1}^{n} (2i - 1) = n^2$$

Pattern. State $P(k)$, add exactly the next term, and close with "by induction".

Counting

Rule	Formula
Permutations	$P(n, r) = \dfrac{n!}{(n-r)!}$
Combinations	$\binom{n}{r} = \dfrac{n!}{r!\,(n-r)!}$
Inclusion–exclusion	$\lvert A \cup B \rvert = \lvert A \rvert + \lvert B \rvert - \lvert A \cap B \rvert$

Trap. Pigeonhole proofs live or die on choosing the holes. Name pigeons and holes explicitly.

Graphs

Euler's formula. For a connected planar graph, $v - e + f = 2$, so $e \leq 3v - 6$ when $v \geq 3$. Since K_5 has $10 > 9$ edges, it is not planar.